# CHEM 1301 Driller 9-3 — Answer Key

**Worked solutions with course-material citations**

Every numeric answer was recomputed by `_scripts/build_driller_9_3.py`, which builds this key and the driller from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …). Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the closest homework screenshot.

**How to read the solutions.** Each calculation is a chain of conversion factors written want-over-have. The label under each factor says where its numbers come from (a molar mass, or two coefficients from the balanced equation), and the crossed-out units are the ones that cancel. Study guide section 9.0 explains this format.

**Score bands.** 14–15 solid on 8.7 and Chapter 9 · 11–13 rework the study guide sections cited for the questions you missed · 10 or below go back through study guide sections 8.7 and 9.2–9.7 before the next driller.

| Q | Ans | Answer | Type | Topic (source) |
|---|---|---|---|---|
| 1 | **e** | divide every element's moles by the smallest mole value | concept | Step 3 of the empirical-formula method (Ch 8 sl. 42–43) |
| 2 | **b** | $C_{12}H_{22}O_{11}$ | calc | Molecular to empirical formula (Ch 8 sl. 39–40) |
| 3 | **a** | $Fe_3O_4$ | calc | Grams to empirical formula (×3) (Ch 8 sl. 42–46) |
| 4 | **e** | $KHCO_3$ | calc | Percent composition to empirical formula (four elements) (Ch 8 sl. 35–37, 42) |
| 5 | **a** | $Mn_2O_7$ | calc | Percent composition to empirical formula (×2) (Ch 8 sl. 42–43) |
| 6 | **b** | $C_5H_{10}O_5$ | calc | Empirical to molecular formula (Ch 8 sl. 47–49) |
| 7 | **d** | 3.00 mol | calc | From percent composition to a mole ratio (Ch 8 sl. 42–49; Ch 9 sl. 9–11) |
| 8 | **e** | Some of it remains unreacted after the reaction stops | concept | The excess reactant (Ch 9 sl. 21–22, 25) |
| 9 | **d** | 3.75 mol | calc | Mole–mole conversion (Ch 9 sl. 9–11) |

| Q | Ans | Answer | Type | Topic (source) |
|---|---|---|---|---|
| 10 | **a** | 0.0501 mol | calc | Grams to moles of another substance (Ch 9 sl. 14, 18) |
| 11 | **d** | 28.0 g | calc | Grams to grams (Ch 9 sl. 14, 19–20) |
| 12 | **c** | 7.47 g | calc | Balance first, then grams to grams (Ch 9 sl. 14, 16, 19) |
| 13 | **c** | C is limiting, and 4.98 g of $SiO_2$ is left over | calc | Limiting reactant and leftover (Ch 9 sl. 21–24) |
| 14 | **b** | 21.0 g | calc | Theoretical yield from two masses (Ch 9 sl. 24–27) |
| 15 | **c** | 84.7% | calc | Percent yield (Ch 9 sl. 28–32) |

Mix: 2 concept / 13 calculation · Ch 8: 6 · Ch 9: 9

## 1. Step 3 of the empirical-formula method — e) divide every element's moles by the smallest mole value

**Source:** Chapter 8 post, slide 42 (Steps for Determining the Empirical Formula, step 3), slides 35–37
**Homework match:** HW21 — hw21ex2.png (empirical formula from percent composition)
**What it tests:** The order of the empirical-formula method, aimed at its most-missed step.

Slide 42's steps, as in study guide 8.7 Type B:

1. Get grams of each element (assume 100 g if given percents).
2. Convert each element's grams to **moles** (÷ its atomic mass).
3. **Divide every mole value by the smallest mole value.** The smallest becomes exactly 1, and the others tell you how many of that atom there are per 1 of the smallest.
4. If any result isn't within 0.1 of a whole number, multiply **every** result by the same small integer.

Why the other choices are wrong:

- a) divide every element's moles by the molar mass of the compound: The molar mass is used later, to get the **molecular** formula (n = molar mass ÷ empirical mass).
- b) divide each element's mass by the smallest mass: Masses aren't atom counts. The ratio has to be in **moles**, and you already converted to moles.
- c) divide every element's moles by the largest mole value: Dividing by the largest makes the results fractions below 1. The smallest makes the smallest element exactly 1.
- d) round each mole value to the nearest whole number: Mole values like 0.0998 or 1.852 aren't subscripts yet. First divide by the smallest to get the ratio.

## 2. Molecular to empirical formula — b) $C_{12}H_{22}O_{11}$

**Source:** Chapter 8 post, slide 39 (empirical formula = smallest whole-number ratio), slide 40
**Homework match:** HW20 — hw20ex2.png (formulas such as theobromine); HW21 — hw21ex3.png

**What it tests:** Type A (study guide 8.7) when the formula is already empirical: the only common divisor is 1.

**Step 1: list the subscripts.** C 12, H 22, O 11.

**Step 2: find the largest number that divides all three evenly.** 11 is prime, so the only candidates are 1 and 11. 12 ÷ 11 isn't whole, so the only common divisor is **1**. (2 divides 12 and 22 but not 11.)

**Step 3: conclusion.** Dividing by 1 changes nothing, so the formula is **already empirical**: $\mathbf{C_{12}H_{22}O_{11}}$.

Why the other choices are wrong:

- a) $C_6H_{11}O_{11}$: Divided only C and H by 2. Every subscript must be divided by the same number.
- c) $C_{12}H_{22}O$: Dropped the O subscript. Every element keeps its count.
- d) $C_6H_{11}O_5$: Divided by 2, but 11 ÷ 2 isn't a whole number (5.5). You can't divide unless **every** subscript divides evenly.
- e) $CH_2O$: That is the empirical formula of glucose ($C_6H_{12}O_6$), not sucrose.

## 3. Grams to empirical formula (×3) — a) $Fe_3O_4$

**Source:** Chapter 8 post, slides 45–46 (empirical formula from masses), slide 43 (step 4: multiply by the smallest integer)
**Homework match:** HW21 — empirical formula from composition data
**What it tests:** The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

**Step 1: grams of each element.** Iron: 1.447 g (given). Oxygen is the rest: 2.000 g − 1.447 g = **0.553 g O**.

**Step 2: grams to moles for each element.** Divide by each element's own atomic mass (periodic table):

$$1.447 \cancel{\text{ g Fe}} \times \underbrace{\frac{1 \text{ mol Fe}}{55.85 \cancel{\text{ g Fe}}}}_{\text{atomic mass of Fe}} = 0.02591 \text{ mol Fe}$$

$$0.553 \cancel{\text{ g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{ g O}}}}_{\text{atomic mass of O}} = 0.03456 \text{ mol O}$$

**Step 3: divide every mole value by the smallest one.** The smallest mole value is 0.02591 mol (Fe). Divide **each** element's moles by it:

| Element | Moles (step 2) | ÷ smallest (0.02591 mol Fe) | × 3 | Subscript |
|---|---|---|---|---|
| Fe | 0.02591 | 1.00 | 3.00 | 3 |
| O | 0.03456 | 1.33 | 4.00 | 4 |

**Step 4: whole numbers?** 1.33 is not within 0.1 of a whole number. It ends in a decimal near .33 or .67 (thirds), so multiply **every** result by 3, including the elements that were already whole (last columns of the table). Empirical formula: $\mathbf{Fe_3O_4}$.

Why the other choices are wrong:

- b) $Fe_6O_8$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- c) FeO: Rounded the fractional result instead of multiplying (only results within 0.1 of a whole number may be rounded).
- d) $Fe_3O$: Used the masses (percents) directly instead of converting to moles first.
- e) $FeO_4$: Multiplied only the fractional element by 3. The same multiplier goes on **every** element.

## 4. Percent composition to empirical formula (four elements) — e) $KHCO_3$

**Source:** Chapter 8 post, slides 35–37 (four-element example: $C_2H_5O_2N$), slide 42
**Homework match:** HW21 — hw21ex2.png (empirical formula from percent composition: $MgSO_4$)
**What it tests:** The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

**Step 1: percent to grams.** Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 39.05 g K, 1.01 g H, 12.00 g C, 47.94 g O.

**Step 2: grams to moles for each element.** Divide by each element's own atomic mass (periodic table):

$$39.05 \cancel{\text{ g K}} \times \underbrace{\frac{1 \text{ mol K}}{39.10 \cancel{\text{ g K}}}}_{\text{atomic mass of K}} = 0.9987 \text{ mol K}$$

$$1.01 \cancel{\text{ g H}} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \cancel{\text{ g H}}}}_{\text{atomic mass of H}} = 1.002 \text{ mol H}$$

$$12.00 \cancel{\text{ g C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \cancel{\text{ g C}}}}_{\text{atomic mass of C}} = 0.9992 \text{ mol C}$$

$$47.94 \cancel{\text{ g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{ g O}}}}_{\text{atomic mass of O}} = 2.996 \text{ mol O}$$

**Step 3: divide every mole value by the smallest one.** The smallest mole value is 0.9987 mol (K). Divide **each** element's moles by it:

| Element | Moles (step 2) | ÷ smallest (0.9987 mol K) | Subscript |
|---|---|---|---|
| K | 0.9987 | 1.00 | 1 |
| H | 1.002 | 1.00 | 1 |
| C | 0.9992 | 1.00 | 1 |
| O | 2.996 | 3.00 | 3 |

**Step 4: whole numbers?** Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: **$KHCO_3$**.

Why the other choices are wrong:

- a) $K_2HCO_3$: Miscounted K: its step-3 result is 1.00, which gives 1, not 2.
- b) $K_{39}HC_{12}O_{47}$: Used the masses (percents) directly instead of converting to moles first.
- c) $K_2H_2C_2O_6$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- d) $K_3HCO$: Subscripts swapped between the elements.

## 5. Percent composition to empirical formula (×2) — a) $Mn_2O_7$

**Source:** Chapter 8 post, slides 42–43 (Steps for Determining the Empirical Formula; step 4: multiply when not whole)
**Homework match:** HW21 — hw21ex2.png (empirical formula from percent composition)
**What it tests:** The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

**Step 1: percent to grams.** Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 49.55 g Mn, 50.45 g O.

**Step 2: grams to moles for each element.** Divide by each element's own atomic mass (periodic table):

$$49.55 \ \cancel{\text{g Mn}} \times \underbrace{\frac{1 \text{ mol Mn}}{54.94 \ \cancel{\text{g Mn}}}}_{\text{atomic mass of Mn}} = 0.9019 \text{ mol Mn}$$

$$50.45 \ \cancel{\text{g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \ \cancel{\text{g O}}}}_{\text{atomic mass of O}} = 3.153 \text{ mol O}$$

**Step 3: divide every mole value by the smallest one.** The smallest mole value is 0.9019 mol (Mn). Divide **each** element's moles by it:

| Element | Moles (step 2) | ÷ smallest (0.9019 mol Mn) | × 2 | Subscript |
|---|---|---|---|---|
| Mn | 0.9019 | 1.00 | 2.00 | 2 |
| O | 3.153 | 3.50 | 6.99 | 7 |

**Step 4: whole numbers?** 3.50 is not within 0.1 of a whole number. It ends in a decimal near .5 (halves), so multiply **every** result by 2, including the elements that were already whole (last columns of the table). Empirical formula: $\mathbf{Mn_2O_7}$.

Why the other choices are wrong:

- b) MnO: Used the masses (percents) directly instead of converting to moles first.
- c) $Mn_4O_{14}$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- d) $MnO_3$: Rounded the fractional result instead of multiplying (only results within 0.1 of a whole number may be rounded).
- e) $MnO_7$: Multiplied only the fractional element by 2. The same multiplier goes on **every** element.

## 6. Empirical to molecular formula — b) $C_5H_{10}O_5$

**Source:** Chapter 8 post, slide 48 (molecular formula = (empirical formula)$_n$), slide 49 (LA Practice), slide 47
**Homework match:** HW21 — hw21ex3.png (empirical formula + molar mass → molecular formula)
**What it tests:** Molecular formula = (empirical formula)$_n$ with n = molar mass ÷ empirical formula mass.

**Step 1: empirical formula mass.** Add up the atomic masses in the empirical formula: $CH_2O$ = 12.01 + 2(1.008) + 16.00 = 30.03 g/mol.

**Step 2: find n.**

$$n = \frac{\text{molar mass of the compound}}{\text{empirical formula mass}} = \frac{150.13 \text{ g/mol}}{30.03 \text{ g/mol}} = 5.00 \approx 5$$

**Step 3: multiply every subscript by n.** $(CH_2O)_5$: C 1 × 5 = 5, H 2 × 5 = 10, O 1 × 5 = 5. Molecular formula: $\mathbf{C_5H_{10}O_5}$.

Why the other choices are wrong:

- a) $CH_2O$: That is the empirical formula itself (n = 1). Its molar mass would be 30.03 g/mol, not 150.13.
- c) $C_5H_2O$: Multiplied only the first subscript by n = 5. **Every** subscript gets multiplied.
- d) $C_4H_8O_4$: n = 4 would give 120.10 g/mol.
- e) $C_6H_{12}O_6$: n = 6 would give a molar mass of 180.16 g/mol.

## 7. From percent composition to a mole ratio — d) 3.00 mol

**Source:** Chapter 8 post, slides 42–49 (empirical then molecular formula); Chapter 9 post, slide 9 (balancing a combustion: double to clear 7/2), slide 11
**Homework match:** HW21 — hw21ex3.png; HW22 — hw22ex1.png ($2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$)
**What it tests:** Section 8.7 feeding Chapter 9: percent composition → empirical → molecular formula → balanced equation → mole ratio.

**Step 1: percent to grams.** Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 92.26 g C, 7.74 g H.

**Step 2: grams to moles for each element.** Divide by each element's own atomic mass (periodic table):

$$92.26 \cancel{\text{ g C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \cancel{\text{ g C}}}}_{\text{atomic mass of C}} = 7.682 \text{ mol C}$$

$$7.74 \cancel{\text{ g H}} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \cancel{\text{ g H}}}}_{\text{atomic mass of H}} = 7.679 \text{ mol H}$$

**Step 3: divide every mole value by the smallest one.** The smallest mole value is 7.679 mol (H). Divide **each** element's moles by it:

| Element | Moles (step 2) | ÷ smallest (7.679 mol H) | Subscript |
|---|---|---|---|
| C | 7.682 | 1.00 | 1 |
| H | 7.679 | 1.00 | 1 |

**Step 4: whole numbers?** Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: **CH**.

**Step 5: molecular formula.** Empirical formula mass: CH = 12.01 + 1.008 = 13.02 g/mol. n = 78.11 ÷ 13.02 = 6.00 ≈ 6, so multiply every subscript by 6: $\mathbf{C_6H_6}$.

**Step 6: write the balanced equation.** Combustion of $C_6H_6$: C first ($6CO_2$), then H ($3H_2O$), then O: 6(2) + 3(1) = 15 O on the right, so 15/2 $O_2$. That isn't whole, so double everything: $\mathbf{2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O}$.

**Step 7: moles $\mathbf{C_6H_6}$ to moles $\mathbf{O_2}$.** Have $C_6H_6$ (coefficient **2**), want $O_2$ (coefficient **15**).

$$0.4000 \cancel{\text{ mol } C_6H_6} \times \underbrace{\frac{15 \text{ mol } O_2}{2 \cancel{\text{ mol } C_6H_6}}}_{\text{coefficient of } O_2 \text{ / coefficient of } C_6H_6} = \boxed{3.00 \text{ mol } O_2}$$

The molecular formula matters here. A mole of the **empirical** unit is a different amount of substance from a mole of the real molecule, so it would give a different number of moles of product.

Why the other choices are wrong:

- a) 0.400 mol: Assumed 1 : 1. The ratio is 15 $O_2$ : 2 $C_6H_6$.
- b) 0.500 mol: Used the empirical formula CH ($CH + 1.25O_2$), which is one-sixth of a molecule.
- c) 2.40 mol: Used $CO_2$'s coefficient (12) instead of $O_2$'s (15).
- e) 6.00 mol: Used 15 but forgot to divide by $C_6H_6$'s coefficient (2).

## 8. The excess reactant — e) Some of it remains unreacted after the reaction stops

**Source:** Chapter 9 post, slide 21 (Limiting Reactants), slide 22, slide 25 (theoretical yield from the limiting reactant)
**Homework match:** HW23 — hw23ex2.png (limiting reactant, then mass of product)
**What it tests:** Roles of the limiting and excess reactants.

The limiting reactant runs out first and fixes how much product forms (slides 21, 25). The other reactant is **in excess**: when the limiting reactant is gone the reaction stops, and **some of the excess reactant is left over**, unreacted.

Why the other choices are wrong:

- a) It is used up before the limiting reactant: Backwards: the limiting reactant runs out first.
- b) It determines the theoretical yield: The **limiting** reactant determines the theoretical yield (slide 25).
- c) Its leftover amount is added to the theoretical yield: Leftover reactant never became product, so it isn't part of the yield.
- d) It is always the reactant present in the larger mass: Mass doesn't decide it. Only a mole comparison through the ratio does.

## 9. Mole–mole conversion — d) 3.75 mol

**Source:** Chapter 9 post, slide 11 (Mole–Mole Conversions Practice 2: $2C_2H_6 + 7O_2$), slide 9
**Homework match:** HW22 — hw22ex1.png ($2C_6H_6 + 15O_2$: reading mole amounts from coefficients)
**What it tests:** mol → mol → mole ratio → mol.

**Step 1: moles $C_8H_{18}$ to moles $O_2$.** Have $C_8H_{18}$ (coefficient **2**), want $O_2$ (coefficient **25**). The question asks for **moles**, so stop here.

$$0.3000 \cancel{\text{ mol } C_8H_{18}} \times \underbrace{\frac{25 \text{ mol } O_2}{2 \cancel{\text{ mol } C_8H_{18}}}}_{\text{coefficient of } O_2 \text{ / coefficient of } C_8H_{18}} = \boxed{3.75 \text{ mol } O_2}$$

Why the other choices are wrong:

- a) 0.0240 mol: Mole ratio upside down: × 2/25 instead of × 25/2.
- b) 0.300 mol: Assumed 1 : 1. The coefficients are 2 ($C_8H_{18}$) and 25 ($O_2$).
- c) 2.40 mol: Used $CO_2$'s coefficient (16) instead of $O_2$'s (25).
- e) 120 mol: Multiplied by the molar mass of $O_2$. That gives grams, not moles.

## 10. Grams to moles of another substance — a) 0.0501 mol

**Source:** Chapter 9 post, slide 18 (Stoichiometry Practice 2: moles of $H_2SO_4$ from 19.9 g KOH), slide 14; Chapter 8 post, slide 43 ($2HgO \rightarrow 2Hg + O_2$)
**Homework match:** HW22 — hw22ex3.png (moles of a substance from grams of another)
**What it tests:** g → mol → mole ratio → mol.

**Step 1: molar masses.** HgO = 200.59 + 16.00 = 216.59 g/mol.

**Step 2: grams HgO to moles HgO.**

$$21.7 \cancel{\text{ g HgO}} \times \underbrace{\frac{1 \text{ mol HgO}}{216.59 \cancel{\text{ g HgO}}}}_{\text{molar mass of HgO}} = 0.1002 \text{ mol HgO}$$

**Step 3: moles HgO to moles $O_2$.** Have HgO (coefficient **2**), want $O_2$ (coefficient **1**). The question asks for **moles**, so stop here.

$$0.1002 \cancel{\text{ mol HgO}} \times \underbrace{\frac{1 \text{ mol } O_2}{2 \cancel{\text{ mol HgO}}}}_{\text{coefficient of } O_2 \text{ / coefficient of HgO}} = \boxed{0.0501 \text{ mol } O_2}$$

Why the other choices are wrong:

- b) 0.100 mol: Assumed 1 : 1. The coefficients are 2 (HgO) and 1 ($O_2$).
- c) 0.200 mol: Mole ratio upside down: × 2/1 instead of × 1/2.
- d) 1.60 mol: Multiplied by the molar mass of $O_2$. That gives grams, not moles.
- e) 4.99 mol: Divided the molar mass by the grams (216.59 ÷ 21.7).

## 11. Grams to grams — d) 28.0 g

**Source:** Chapter 9 post, slides 19–20 (Stoichiometry Practice: g → mol → mol → g), slide 14
**Homework match:** HW23 — hw23ex1.png (grams of product from grams of reactant)
**What it tests:** g → mol → mole ratio → g.

**Step 1: molar masses.** $Fe_2O_3 = 2(55.85) + 3(16.00) = 159.70$ g/mol; Fe = 55.85 g/mol (periodic table).

**Step 2: grams $Fe_2O_3$ to moles $Fe_2O_3$.**

$$40.0 \text{ g } \cancel{Fe_2O_3} \times \underbrace{\frac{1 \text{ mol } Fe_2O_3}{159.70 \text{ g } \cancel{Fe_2O_3}}}_{\text{molar mass of } Fe_2O_3} = 0.2505 \text{ mol } Fe_2O_3$$

**Step 3: moles $Fe_2O_3$ to moles Fe.** Have $Fe_2O_3$ (coefficient **1**), want Fe (coefficient **2**).

$$0.2505 \text{ } \cancel{\text{mol } Fe_2O_3} \times \underbrace{\frac{2 \text{ mol Fe}}{1 \text{ } \cancel{\text{mol } Fe_2O_3}}}_{\text{coefficient of Fe / coefficient of } Fe_2O_3} = 0.5009 \text{ mol Fe}$$

**Step 4: moles Fe to grams Fe.**

$$0.5009 \text{ } \cancel{\text{mol Fe}} \times \underbrace{\frac{55.85 \text{ g Fe}}{1 \text{ } \cancel{\text{mol Fe}}}}_{\text{molar mass of Fe}} = \boxed{28.0 \text{ g Fe}}$$

Why the other choices are wrong:

- a) 0.501 g: That is the **moles** of Fe. The last step (× molar mass) is missing.
- b) 6.99 g: Mole ratio upside down: × 1/2 instead of × 2/1.
- c) 14.0 g: Assumed 1 : 1. The coefficients are 1 ($Fe_2O_3$) and 2 (Fe).
- e) 80.0 g: Used the molar mass of $Fe_2O_3$ in the last step instead of Fe's.

## 12. Balance first, then grams to grams — c) 7.47 g

**Source:** Chapter 9 post, slide 16 (balance before using coefficients), slide 14, slide 19
**Homework match:** HW22 — hw22ex1.png (balance with the lowest whole-number coefficients); HW23 — hw23ex3.png
**What it tests:** g → mol → mole ratio → g, after balancing.

**Step 1: balance the equation.** $Li_2O$ has 2 Li and 1 O. $O_2$ supplies 2 O, so make **2**$Li_2O$. That needs 4 Li: $\mathbf{4Li + O_2 \rightarrow 2Li_2O}$.

**Step 2: molar masses.** Li = 6.94 g/mol (periodic table); $Li_2O = 2(6.94) + 16.00 = 29.88$ g/mol.

**Step 3: grams Li to moles Li.**

$$3.47 \text{ } \cancel{\text{g Li}} \times \underbrace{\frac{1 \text{ mol Li}}{6.94 \text{ } \cancel{\text{g Li}}}}_{\text{molar mass of Li}} = 0.5000 \text{ mol Li}$$

**Step 4: moles Li to moles $Li_2O$.** Have Li (coefficient **4**), want $Li_2O$ (coefficient **2**).

$$0.5000 \text{ mol Li} \times \underbrace{\frac{2 \text{ mol } Li_2O}{4 \text{ mol Li}}}_{\text{coefficient of } Li_2O \text{ / coefficient of Li}} = 0.2500 \text{ mol } Li_2O$$

**Step 5: moles $Li_2O$ to grams $Li_2O$.**

$$0.2500 \text{ mol } Li_2O \times \underbrace{\frac{29.88 \text{ g } Li_2O}{1 \text{ mol } Li_2O}}_{\text{molar mass of } Li_2O} = \boxed{7.47 \text{ g } Li_2O}$$

Why the other choices are wrong:

- a) 0.250 g: That is the **moles** of $Li_2O$. The last step (× molar mass) is missing.
- b) 1.74 g: Used the molar mass of Li in the last step instead of $Li_2O$'s.
- d) 14.9 g: Assumed 1 : 1. The coefficients are 4 (Li) and 2 ($Li_2O$).
- e) 29.9 g: Mole ratio upside down: × 4/2 instead of × 2/4.

## 13. Limiting reactant and leftover — c) C is limiting, and 4.98 g of $SiO_2$ is left over

**Source:** Chapter 9 post, slide 24 (limiting reactant from two masses), slides 21–23
**Homework match:** HW23 — hw23ex2.png (determine the limiting reactant)
**What it tests:** Limiting reactant by need-vs-have in moles, then the leftover mass of the excess reactant.

**Step 1: molar masses.** $SiO_2 = 28.09 + 2(16.00) = 60.09$ g/mol; C = 12.01 g/mol (periodic table).

**Step 2: grams to moles for each reactant.**

$$30.0 \text{ g } SiO_2 \times \underbrace{\frac{1 \text{ mol } SiO_2}{60.09 \text{ g } SiO_2}}_{\text{molar mass of } SiO_2} = 0.4993 \text{ mol } SiO_2$$

$$15.0 \text{ g C} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \text{ g C}}}_{\text{molar mass of C}} = 1.249 \text{ mol C}$$

**Step 3: how much C does all the $SiO_2$ need?** Have $SiO_2$ (coefficient **1**), want C (coefficient **3**).

$$0.4993 \text{ mol } SiO_2 \times \underbrace{\frac{3 \text{ mol C}}{1 \text{ mol } SiO_2}}_{\text{coefficient of C / coefficient of } SiO_2} = 1.498 \text{ mol C}$$

**Step 4: compare.** You **need** 1.498 mol C but only **have** 1.249 mol, so **C is limiting** and $SiO_2$ is in excess.

**Step 5: how much $SiO_2$ reacts?** All 1.249 mol of the limiting C reacts. Have C (coefficient **3**), want $SiO_2$ (coefficient **1**).

$$1.249 \text{ mol C} \times \underbrace{\frac{1 \text{ mol } SiO_2}{3 \text{ mol C}}}_{\text{coefficient of } SiO_2 \text{ / coefficient of C}} = 0.4163 \text{ mol } SiO_2$$

$$0.4163 \cancel{\text{mol SiO}_2} \times \underbrace{\frac{60.09 \text{ g SiO}_2}{1 \cancel{\text{mol SiO}_2}}}_{\text{molar mass of SiO}_2} = 25.0 \text{ g SiO}_2$$

**Step 6: left over.** 30.0 g $SiO_2$ at the start − 25.0 g that reacts = **4.98 g left over**.

Mass alone never decides the limiting reactant: compare in moles, through the mole ratio.

Why the other choices are wrong:

- a) C is limiting, and 30.0 g of $SiO_2$ is left over: That is **all** of the $SiO_2$. Some of it reacts with the C, so less is left.
- b) C is limiting, and 25.0 g of $SiO_2$ is left over: That is the mass of $SiO_2$ that **reacts**. Left over = 30.0 g − 25.0 g.
- d) Neither is limiting; both reactants are used up completely: That needs the moles to be in exactly a 1 : 3 ratio, and they aren't.
- e) $SiO_2$ is limiting, and C is left over: Check what all of one reactant needs. The need-vs-have comparison shows the C runs out first.

## 14. Theoretical yield from two masses — b) 21.0 g

**Source:** Chapter 9 post, slide 27 (theoretical yield when both masses are given), slide 25, slide 24
**Homework match:** HW24 — hw24ex2.png (theoretical yield of $XeF_4$ from two masses)
**What it tests:** Moles of each reactant → limiting reactant → product from the limiting reactant only.

**Step 1: molar masses.** $NO_2 = 14.01 + 2(16.00) = 46.01$ g/mol; $H_2O = 2(1.008) + 16.00 = 18.02$ g/mol; $HNO_3 = 1.008 + 14.01 + 3(16.00) = 63.02$ g/mol.

**Step 2: grams to moles for each reactant.**

$$23.0 \cancel{\text{g NO}_2} \times \underbrace{\frac{1 \text{ mol NO}_2}{46.01 \cancel{\text{g NO}_2}}}_{\text{molar mass of NO}_2} = 0.4999 \text{ mol NO}_2$$

$$5.00 \cancel{\text{g H}_2\text{O}} \times \underbrace{\frac{1 \text{ mol H}_2\text{O}}{18.02 \cancel{\text{g H}_2\text{O}}}}_{\text{molar mass of H}_2\text{O}} = 0.2775 \text{ mol H}_2\text{O}$$

**Step 3: how much $H_2O$ does all the $NO_2$ need?** Have $NO_2$ (coefficient **3**), want $H_2O$ (coefficient **1**).

$$0.4999 \cancel{\text{mol NO}_2} \times \underbrace{\frac{1 \text{ mol H}_2\text{O}}{3 \cancel{\text{mol NO}_2}}}_{\text{coefficient of H}_2\text{O} \,/\, \text{coefficient of NO}_2} = 0.1666 \text{ mol H}_2\text{O needed}$$

**Step 4: compare.** You **have** 0.2775 mol $H_2O$ and only **need** 0.1666 mol. $H_2O$ is in excess, so **$NO_2$ is limiting**.

**Step 5: theoretical yield from the limiting reactant ($NO_2$).** Have $NO_2$ (coefficient **3**), want $HNO_3$ (coefficient **2**).

$$0.4999 \text{ mol } \cancel{NO_2} \times \underbrace{\frac{2 \text{ mol } HNO_3}{3 \text{ mol } \cancel{NO_2}}}_{\text{coefficient of } HNO_3 \text{ / coefficient of } NO_2} = 0.3333 \text{ mol } HNO_3$$

$$0.3333 \text{ mol } \cancel{HNO_3} \times \underbrace{\frac{63.02 \text{ g } HNO_3}{1 \text{ mol } \cancel{HNO_3}}}_{\text{molar mass of } HNO_3} = \boxed{21.0 \text{ g } HNO_3}$$

**Cross-check:** all the $H_2O$ would make 35.0 g $HNO_3$. The smaller amount, 21.0 g, is the theoretical yield.

Why the other choices are wrong:

- a) 0.333 g: That is the **moles** of $HNO_3$.
- c) 28.0 g: Added the reactant masses. Part of the excess reactant is left over.
- d) 31.5 g: Used $NO_2$ but assumed 1 : 1. The ratio is 2 $HNO_3$ : 3 $NO_2$.
- e) 35.0 g: Calculated from $H_2O$. $H_2O$ is in excess, so this is more than the $NO_2$ can make.

## 15. Percent yield — c) 84.7%

**Source:** Chapter 9 post, slide 30 (percent yield formula), slides 31–32 (worked percent yields), slide 28
**Homework match:** HW24 — hw24ex1.png (percent yield)
**What it tests:** Two parts: theoretical yield from the limiting reactant, then actual / theoretical × 100.

**Step 1: which reactant controls the yield?** $C_6H_{12}O_6$ is the only reactant, so it is limiting.

**Step 2: molar masses.** $C_6H_{12}O_6 = 6(12.01) + 12(1.008) + 6(16.00) = 180.16$ g/mol; $C_2H_5OH = 2(12.01) + 6(1.008) + 16.00 = 46.07$ g/mol.

**Step 3: theoretical yield of $C_2H_5OH$.** Have $C_6H_{12}O_6$ (coefficient **1**), want $C_2H_5OH$ (coefficient **2**).

$$45.0 \text{ g } \cancel{C_6H_{12}O_6} \times \underbrace{\frac{1 \text{ mol } C_6H_{12}O_6}{180.16 \text{ g } \cancel{C_6H_{12}O_6}}}_{\text{molar mass of } C_6H_{12}O_6} = 0.2498 \text{ mol } C_6H_{12}O_6$$

$$0.2498 \text{ mol } \cancel{C_6H_{12}O_6} \times \underbrace{\frac{2 \text{ mol } C_2H_5OH}{1 \text{ mol } \cancel{C_6H_{12}O_6}}}_{\text{coefficient of } C_2H_5OH \text{ / coefficient of } C_6H_{12}O_6} = 0.4996 \text{ mol } C_2H_5OH$$

$$0.4996 \text{ mol } \cancel{C_2H_5OH} \times \underbrace{\frac{46.07 \text{ g } C_2H_5OH}{1 \text{ mol } \cancel{C_2H_5OH}}}_{\text{molar mass of } C_2H_5OH} = 23.01 \text{ g } C_2H_5OH \text{ (theoretical)}$$

**Step 4: percent yield.** The actual yield (19.5 g, measured) goes on top, the theoretical yield on the bottom:

$$\text{percent yield} = \frac{19.5 \text{ g (actual)}}{23.01 \text{ g (theoretical)}} \times 100\% = \boxed{84.7\%}$$

Why the other choices are wrong:

- a) 0.847%: Forgot to multiply by 100.
- b) 43.3%: Divided by the mass of $C_6H_{12}O_6$. The bottom must be the theoretical yield of $C_2H_5OH$.
- d) 118%: Upside down: theoretical / actual. A yield can't be over 100%.
- e) 169%: Theoretical yield found with a 1 : 1 ratio. The coefficients are 1 ($C_6H_{12}O_6$) and 2 ($C_2H_5OH$).